Carnegie Mellon University
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Thesis Proposal

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Simulation-based Time Series Analysis: Likelihood-free Estimation, Testing for Structure, and Prediction with Transformers using Synthetic Data

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Abstract

Technological advances have enabled the collection of large-scale time series data in many fields. Traditional time series methods often struggle to capture the complex dynamics present in such data because of their restrictive assumptions, such as stationarity, linear dynamics, or Gaussian noise. Consequently, there is growing demand for more flexible time series methods. At the same time, recent developments in machine learning, notably the transformer architecture, have demonstrated remarkable success in sequence modeling. However, the statistical foundations of transformer-based methods for time series remain poorly understood, particularly in nonstationary settings.

In this thesis proposal, we present simulation-based methods for problems in multivariate time series analysis. In the first part, we introduce a bootstrap-based framework for conditional independence testing with nonstationary time series that utilizes nonlinear regression. In the second part, we develop a framework for simulation-based parameter estimation based on matching random features, which we apply to nonlinear non-Gaussian state-space models, discrete-time dynamical systems and ODEs with observational noise, discretized Lévy-driven SDEs, and more. In the third part, we introduce a simulation-and-regression framework for time series prediction with transformers that accounts for parameter uncertainty.

Contents

1 Introduction

1.1 Thesis Agenda

We provide a concise overview of the organization of this thesis proposal, along with some motivations for each topic.

I. Conditional Independence Testing (Section 2). Given nonstationary time series $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, we develop a nonlinear regression-based test of the null hypothesis that X_t is conditionally independent of Y_t given Z_t at all times t, with any desired leads and lags incorporated into these variables. Tests of conditional independence are used in many problems, such as variable selection, causal discovery, the validation of theory-based causal graphs, and the evaluation of predictor fairness; see Hardt et al. (2016).

II. Simulation-based Estimation (Section 3). Next, we introduce a simulation-based framework for estimating the unknown parameter in a parametric statistical model for a time series $X_{1:n}$. Such methods are desirable for models where the likelihood is analytically or computationally intractable, but simulating from the model is easy. For example, nonlinear non-Gaussian state-space models, discrete-time dynamical systems and ODEs with observational noise, and discretized Lévy-driven SDEs.

III. Prediction with Transformers (Section 4). Lastly, we develop a simulation-and-regression framework for training transformers for the tasks of forecasting and smoothing. Forecasting aims to characterize the future values of a stochastic process given the current information. It underpins monetary policy decisions in economics, portfolio optimization in finance, public health policy decisions during epidemics, and early warning systems for extreme weather events. Smoothing aims to characterize the past values of a latent stochastic process given the current information. Applications include estimating sentiment in the social sciences, stochastic volatility in finance, infectivity in epidemiology, and atmospheric variables in the Earth Sciences.

2 Conditional Independence Testing

2.1 Overview

In this part of the thesis, we introduce a framework for conditional independence testing with nonstationary time series. To the best of our knowledge, this is the first conditional independence testing framework for nonstationary time series outside of the linear-Gaussian setting. In particular, we use a theoretical framework for nonstationary time series based on the foundational work of Wu (2005); Zhou and Wu (2009); Dahlhaus et al. (2019). We propose a bootstrap-based testing procedure, which we justify with a distribution-uniform version of the strong Gaussian approximation from Mies and Steland (2023). We prove that our tests have Type I error control, *uniformly* over a large collection of null distributions. The main ideas are presented here, and we refer readers to the paper (Wieck-Sosa et al., 2025) for more details.

2.2 Setting

Let $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$ be time series taking values in $\mathbb{R}^{d_X}$, $\mathbb{R}^{d_Y}$, $\mathbb{R}^{d_Z}$, respectively, for some $n, d_X, d_Y, d_Z \in \mathbb{N}$. To simplify the notation, let these variables incorporate any desired leads and lags. The only requirement is that the conditioning set Z_t is known at time t. In this section, we present a test for the null hypothesis that

$$X_t \perp\!\!\!\perp Y_t \mid Z_t \text{ for all } t \in [n]. \tag{1}$$

We allow the time series to have nonlinear temporal dynamics and general forms of nonstationarity. The paper (Wieck-Sosa et al., 2025) includes the details about the asymptotic framework, the assumptions about the nonstationarity and decay of temporal dependence using the physical dependence measure of Wu (2005), and the rates at which the number of dimensions and time-offsets can grow with n.

2.3 Time-varying Regression and Error Processes

We can always decompose

$$X_t = f_t(Z_t) + \varepsilon_t, \quad Y_t = g_t(Z_t) + \xi_t, \tag{2}$$

where $f_t(z) = \mathbb{E}(X_t \mid Z_t = z)$ and $g_t(z) = \mathbb{E}(Y_t \mid Z_t = z)$ are the regression functions, which may vary over time. The error processes $\varepsilon_{1:n}$ and $\xi_{1:n}$ can also be nonstationary and temporally dependent. Denote the vectorized outer product of the errors at time t by

$$R_t = \mathrm{vec}(\varepsilon_t \xi_t^\top). \tag{3}$$

Next, let $\hat{f}_t$ and $\hat{g}_t$ be estimates of f_t and g_t, respectively, and let

$$\hat{\varepsilon}_t = X_t - \hat{f}_t(Z_t), \quad \hat{\xi}_t = Y_t - \hat{g}_t(Z_t), \tag{4}$$

be the corresponding residuals. Denote the vectorized outer product of these residuals at time t by

$$\hat{R}_t = \mathrm{vec}(\hat{\varepsilon}_t \hat{\xi}_t^\top). \tag{5}$$

Crucially, our testing framework can be used with *any regression estimator*. The required rates of convergence for the regression estimators are discussed in the paper (Wieck-Sosa et al., 2025). In our experiments, we use the sieve estimator from Ding and Zhou (2021), though there are many other regression estimators for nonstationary time series to choose from (Vogt, 2012; Zhang and Wu, 2015; Yousuf and Ng, 2021; Chen et al., 2022; Kurisu et al., 2025). Our test possesses a property known as *rate double robustness*, which means that we only require modest convergence rates for the products of the estimation errors, rather than for each estimation error individually.

2.4 Testing Procedure

Our proposed testing procedure can be viewed as an extension of the *generalized covariance measure* (GCM) test from Shah and Peters (2020) from the iid setting to the nonstationary time series setting, so we refer to it as the *dynamic generalized covariance measure* (dGCM) test. The core idea is that, under the null hypothesis of conditional independence from (1), the covariance of the errors from (2) will be zero at all times. Note that these covariances can be zero at all times, even under alternatives in which the corresponding conditional dependencies always hold. Consequently, we can only hope to have power against alternatives in which these covariances are non-zero for at least *some* times.

Roughly speaking, our test statistic uses the empirical covariances among the residuals from (4) to try to detect non-zero covariances among the errors from (2), for at least *some* times. To estimate the desired quantile of the test statistic, we use a bootstrap procedure justified by a *distribution-uniform* version of the strong Gaussian approximation from Mies and Steland (2023) applied to the process of error products R_t from (3). This is possible because, under the null hypothesis from (1), the error products R_t from (3) will have mean zero at all times t. The approximating Gaussian process has a time-varying covariance structure, which we estimate using a rolling-window approach; see the paper (Wieck-Sosa et al., 2025) for how to select the window size. We prove that the following procedure has Type I error control, uniformly over a large collection of null distributions.

Dynamic Generalized Covariance Measure (dGCM) Test.

- **Inputs:** Data $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, significance level α, number of simulations s, $p \in [2, \infty]$ for norm in test statistic, and method for selecting the window size L.
- Regress X_t on Z_t, $t \in [n]$, and obtain residuals $\hat{\varepsilon}_t$, $t \in [n]$.
- Regress Y_t on Z_t, $t \in [n]$, and obtain residuals $\hat{\xi}_t$, $t \in [n]$.
- Obtain residual products $\hat{R}_t = \mathrm{vec}(\hat{\varepsilon}_t \hat{\xi}_t^\top)$, $t \in [n]$.
- Calculate test statistic on time series of residual products
$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p .$$
- Select the window size L for covariance estimation using $\hat{R}_{1:n}$.
- For $t \in [n]$, obtain estimates of the time-varying covariances as
$$\hat{\Sigma}_t = \frac{1}{L} \left(\sum_{j=t-L+1}^{t} \hat{R}_j \right) \left(\sum_{j=t-L+1}^{t} \hat{R}_j \right)^\top .$$
- For each $r \in [s]$ and $t \in [n]$, simulate *independent* Gaussians
$$\tilde{R}_t^{(r)} \sim N(0, \hat{\Sigma}_t).$$
- For each $r \in [s]$, calculate test statistic on simulated Gaussian process
$$T(\tilde{R}_{1:n}^{(r)}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \tilde{R}_t^{(r)} \right\|_p .$$
- Calculate the $1 - \alpha$ empirical quantile $\hat{q}_{1-\alpha}^{\text{boot}}$ of the simulated test statistics $T(\tilde{R}_{1:n}^{(1)}), \ldots, T(\tilde{R}_{1:n}^{(s)})$.
- Reject null hypothesis from (1) at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\text{boot}}$, else retain.

3 Simulation-based Estimation

3.1 Overview

In this part of the thesis, we develop a simulation-based estimation framework for parametric models of stochastic processes. Given an $\mathbb{R}^d$-valued time series $X_{1:n}$ generated by such a model, we aim to estimate the unknown parameter $\theta_0 \in \mathbb{R}^p$ for the model. Unfortunately, the likelihood function is intractable, but simulating from the model with different parameter values is straightforward.

Our core idea is to estimate θ_0 by tuning the parameter values so that a small number of *random features* of the simulated data closely match those of the observed data. Results from the "geometry from a time series" literature in nonlinear dynamics indicate that models with a p-dimensional parameter space can typically be identified using only $2p+1$ random features, regardless of the dimension of the observations. Our approach avoids both user-selected summary statistics and costly neural network-based procedures for learning them. This leads to a user-friendly method that can be readily applied in many fields, such as ecology and epidemiology.

We introduce two estimators. First, a time-average estimator for processes that are stationary, or at least asymptotically mean stationary. Second, a rolling-window estimator for processes with general forms of nonstationarity. We prove that both estimators are consistent and asymptotically normal under nonrestrictive conditions. Specifically, we use a theoretical framework for nonstationary time series based on Wu (2005); Zhou and Wu (2009); Dahlhaus et al. (2019); Mies and Steland (2023). Our experiments suggest that simulation-based estimation can be made more automatic by matching random features, with little loss in statistical efficiency.

3.2 Notation and setting

For real numbers $x, y \in \mathbb{R}$, denote $x \wedge y = \min(x, y)$ and $x \vee y = \max(x, y)$. For a vector $x \in \mathbb{R}^d$, denote the ℓ^p norm by $\|x\|_p$ and the Euclidean norm by $\|x\| = \|x\|_2$. For a random vector X, $\|X\|_{\mathcal{L}^q} = (\mathbb{E}\|X\|^q)^{1/q}$ denotes the $\mathcal{L}^q$ norm of the Euclidean norm. For any natural number $j \in \mathbb{N}$, denote $[j] = \{1, 2, \ldots, j\}$. For a sequence of random vectors X_t, $t = 1, \ldots, n$, each taking values in $\mathbb{R}^d$, we denote a subsequence as $X_{t_1:t_2} = (X_{t_1}, \ldots, X_{t_2})^\top$, which takes values in $\mathbb{R}^{(t_2-t_1+1)\times d}$ for some $t_1, t_2 \in [n]$.

We observe a time series X_t^{obs}, $t = 1, \ldots, n$, taking values in $\mathbb{R}^d$ for some fixed dimension $d \in \mathbb{N}$. The observed time series is assumed to arise from a known generative model with an unknown p-dimensional parameter $\theta_0 \in \Theta \subset \mathbb{R}^p$, $p \in \mathbb{N}$. Each value of $\theta \in \Theta$ determines a law of a stochastic process. For any value of θ, we can use the generative model to simulate $s \in \mathbb{N}$ realizations of a length n time series $(X_{1:n}^{(r)}(\theta))_{r \in [s]}$. When we do not need to refer to a particular realization, we drop the superscript.

3.3 Random Fourier features

The central idea is that we should estimate the p-dimensional parameter θ_0 by matching the values of a small number of random features of the simulated and observed data. Consider $k = 2p + 1$ randomly drawn functions $\varphi_1, \ldots, \varphi_k$ from a distribution D over a suitably nice function class $\mathcal{F}$. The k functions should be sampled independently of one another and of the data. Each function $\varphi_i : \mathbb{R}^{(m+1)\times d} \to \mathbb{R}$ takes as input a length $m+1$ subsequence of the d-dimensional time series, where m is chosen based on the dynamics of the generative model. For instance, if the model is an order m^* Markov process, then set $m = m^*$.

Let $x_1, \ldots, x_{m+1}$ be vectors in $\mathbb{R}^d$, and define $x = (x_1, \ldots, x_{m+1})^\top \in \mathbb{R}^{(m+1)\times d}$. In this paper, we focus on random Fourier features, though our framework can be used with different random features. Specifically, we consider random Fourier features of the form

$$\varphi_i(x) = \cos\left(\sum_{j=1}^{m+1} \Omega_{i,j} \cdot x_j + \alpha_i\right), \tag{6}$$

where $\Omega_{i,j} \overset{\text{iid}}{\sim} N(0, I_d)$ and $\alpha_i \overset{\text{iid}}{\sim} U(-\pi, \pi)$ for $i = 1, \ldots, k$ and $j = 1, \ldots, m+1$. Denote all k of these functions by

$$\varphi = (\varphi_1, \ldots, \varphi_k), \tag{7}$$

so that $\varphi : \mathbb{R}^{(m+1)\times d} \to \mathbb{R}^k$. Define the i-th random feature of the observed and r-th simulated time series at time $t = m+1, \ldots, n$ as

$$f_{t,i}^{\text{obs}} = \varphi_i(X_{t-m:t}^{\text{obs}}), \quad f_{t,i}^{(r)}(\theta) = \varphi_i\left(X_{t-m:t}^{(r)}(\theta)\right).$$

Write all k random features at time t as

$$f_t^{\text{obs}} = \left(f_{t,1}^{\text{obs}}, \ldots, f_{t,k}^{\text{obs}}\right), \quad f_t^{(r)}(\theta) = \left(f_{t,1}^{(r)}(\theta), \ldots, f_{t,k}^{(r)}(\theta)\right). \tag{8}$$

3.4 Estimators

Time-average estimator. For processes that are asymptotically mean stationary, it suffices to consider the time-average of $k = 2p + 1$ random features. Define the observed and simulated time-averages of the random features by

$$F^{\text{obs}} = \frac{1}{n-m} \sum_{t=m+1}^{n} f_t^{\text{obs}}, \quad \bar{F}^{\text{sim}}(\theta) = \frac{1}{n-m} \sum_{t=m+1}^{n} \bar{f}_t^{\text{sim}}(\theta), \tag{9}$$

where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$. The time-average estimator is given by

$$\hat{\theta}^{\text{TA}} = \operatorname*{argmin}_{\theta\in\Theta} \left\| \hat{Q}_n^{\text{TA}}(\theta) \right\|, \tag{10}$$

where the sample discrepancy function is defined as

$$\hat{Q}_n^{\text{TA}}(\theta) = F^{\text{obs}} - \bar{F}^{\text{sim}}(\theta). \tag{11}$$

In practice, an optimization procedure is used to find the minimizer of $\left\| \hat{Q}_n^{\text{TA}}(\cdot) \right\|$.

Rolling-window estimator. For many nonstationary processes, a different approach is required because the time-average of the time-varying means of the random features may not converge to a limiting mean. Even when it does converge, the limiting mean may not uniquely identify each θ. Thus, we introduce the following estimator.

Define the observed and simulated rolling-window random features at time t by

$$F_t^{\text{obs}} = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} f_j^{\text{obs}}, \quad \bar{F}_t^{\text{sim}}(\theta) = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} \bar{f}_j^{\text{sim}}(\theta), \tag{12}$$

where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$ and $w \in \mathbb{N}$ is the window size. The rolling-window estimator is given by

$$\hat{\theta}^{\text{RW}} = \operatorname*{argmin}_{\theta\in\Theta} \left| \hat{Q}_n^{\text{RW}}(\theta) \right|, \tag{13}$$

where the sample discrepancy function is defined as

$$\begin{aligned} \hat{Q}_n^{\text{RW}}(\theta) &= \frac{1}{n-m} \sum_{t=m+\tau+L}^{n} \left[\left\| F_t^{\text{obs}} - \bar{F}_t^{\text{sim}}(\theta) \right\|^2 + K_t(\theta) \right], \\ K_t(\theta) &= 2\left(F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta)\right)^\top \left(\left[f_t^{\text{obs}} - \bar{f}_t^{\text{sim}}(\theta) \right] - \left[F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta) \right] \right), \end{aligned} \tag{14}$$

where $\tau \in \mathbb{N}$ is an initial time-offset and $L \in \mathbb{N}$ is a lag. In practice, an optimization procedure is used to find the minimizer of $\left|\hat{Q}_n^{\mathrm{RW}}(\cdot)\right|$. We discuss the selection of w, τ, and L in the paper.

Extensions. Several extensions are possible. First, both estimators can be generalized by replacing the Euclidean norm with a weighted norm, using a weight matrix given by the inverse of a suitable estimator of the long-run covariance matrix; see Gourieroux and Monfort (1996). Second, the sample averages used in both estimators can be replaced with different mean estimators. Third, one may extend both estimators to allow for early stopping, so that only a fraction of the time series is used.

3.5 Examples

We assess the finite-sample performance of the proposed estimators on several data generating processes. We optimize the objective functions using the differential evolution solver `differential_evolution` from the `scipy.optimize` module in the SciPy Python library, although our experiments indicate that other optimizers perform comparably. For the ODE examples, we used the Runge–Kutta 5(4) solver `RK45` from the `scipy.integrate` module in SciPy. Each density plot is constructed from $1,000$ independent estimates of the p-dimensional parameter using $2p+1$ random Fourier features and 10 simulations per parameter value. To be clear, each parameter estimate is based on an independent realization of the process and an independent draw of $2p+1$ random features.

Gaussian model. For $t = 1, \ldots, n$, we observe

$$X_t = \mu + \epsilon_t,$$

where $\epsilon_t \overset{\text{iid}}{\sim} N(0,1)$. The unknown parameter is μ, so we use $2p+1 = 3$ random features.

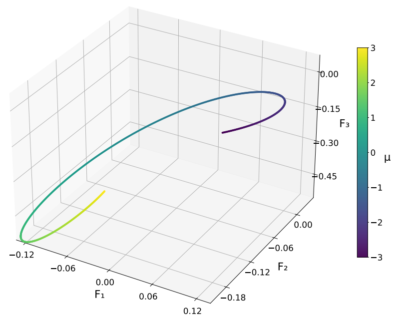
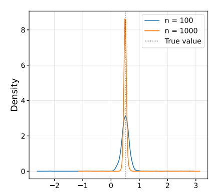


Figure 1: Three random features averaged over $n = 1000$ times and $s = 10$ simulations, for each parameter value on a grid in $[-3, 3]$. Color denotes μ.

We aim to estimate

$$\mu_0 = 0.5,$$

using the time-average estimator.



Logistic map model. For $t = 1, \ldots, n$, define the state by

$$Z_t = \rho Z_{t-1}(1 - Z_{t-1}).$$

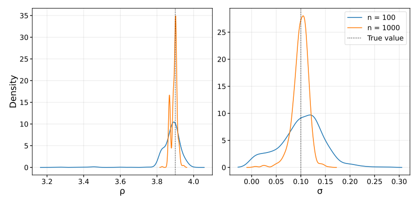
We observe

$$X_t = Z_t + \sigma \epsilon_t,$$

where $\epsilon_t \overset{\text{iid}}{\sim} N(0, 1)$ and the unknown initial value Z_0 is sampled from a $U[0, 1]$ distribution, so that it is fixed before running the procedure. Our goal is to estimate

$$\rho_0 = 3.9, \quad \sigma_0 = 0.1,$$

using the time-average estimator. We emphasize that the logistic map is in a chaotic regime when $\rho = 3.9$; see Devaney (2018) for more information.



State-space model. For times $t = 1, \dots, n$, define the state by

$$Z_t = Z_{t-1} + \psi(\mu - Z_{t-1}) + \sigma \epsilon_t,$$

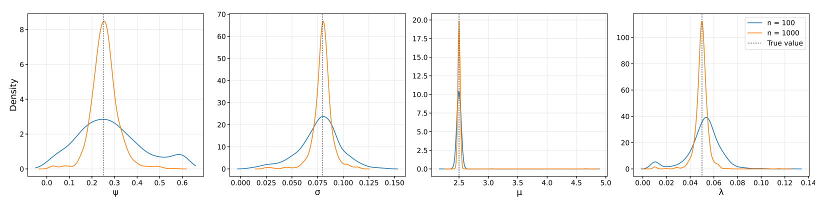
where $\epsilon_t \overset{\text{iid}}{\sim} N(0,1)$. For dimensions $j = 1, \dots, d \equiv 25$, we observe

$$X_t^{(j)} = Z_t + \lambda \xi_t^{(j)},$$

where each $\xi_t^{(j)} \overset{\text{iid}}{\sim} N(0,1)$. We aim to estimate

$$\psi_0 = 0.25, \quad \sigma_0 = 0.08, \quad \mu_0 = 2.5, \quad \lambda_0 = 0.05,$$

using the time-average estimator.



SIR model. For time horizon $T = 50$, let $(S_v, I_v, R_v)_{v \in [0,T]}$, be defined by the susceptible-infected-recovered (SIR) system of ODEs

$$\begin{aligned}
\frac{\mathrm{d}S_v}{\mathrm{d}v} &= -\frac{\beta}{N} S_v \, I_v, \\
\frac{\mathrm{d}I_v}{\mathrm{d}v} &= \frac{\beta}{N} S_v \, I_v - \gamma I_v, \\
\frac{\mathrm{d}R_v}{\mathrm{d}v} &= \gamma I_v,
\end{aligned}$$

where S_v/N, I_v/N, and R_v/N denote the fractions of the population that are susceptible, infected, and recovered (or removed), respectively, $\beta > 0$ is the transmission rate, and $\gamma > 0$ is the recovery rate. Let the known initial values be $S_0 = 980,000$, $I_0 = 20,000$, $R_0 = 0$, where the number of individuals is $N = S_0 + I_0 + R_0 = 1,000,000$.

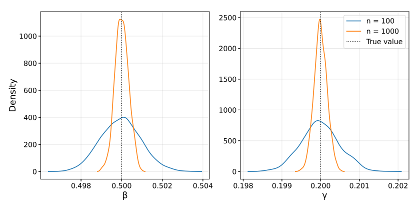
For each $t = 1, \dots, n$, we test $\mathrm{TE}_t = \lfloor N/100 \rfloor$ people with equal probability using a test with sensitivity $\mathrm{SE}_t = 0.95$ and specificity $\mathrm{SP}_t = 0.99$. We observe the proportion of positive tests

$$X_t = Z_t/\mathrm{TE}_t, \quad Z_t \sim \mathrm{Binomial}\left(\mathrm{TE}_t, \pi_t\right),$$

where $\pi_t = \mathrm{SE}_t I_{Tt/n}/N + (1 - \mathrm{SP}_t)\left(1 - I_{Tt/n}/N\right)$. This model allows the number of tests TE_t, sensitivity SE_t, and specificity SP_t to change with t, although we keep them constant for simplicity. We aim to estimate

$$\beta_0 = 0.5, \quad \gamma_0 = 0.2,$$

using the rolling-window estimator. Note that the bounds for the optimizer were set to $[0.01, 1]$ for β and $[0.01, 0.5]$ for γ.



Lotka–Volterra model. For time horizon $T = 20$, let $(Z_v^{(1)}, Z_v^{(2)})_{v \in [0,T]}$ be defined by the Lotka–Volterra predator–prey system of ODEs

$$\begin{aligned} \frac{\mathrm{d}Z_v^{(1)}}{\mathrm{d}v} &= \alpha Z_v^{(1)} - \beta Z_v^{(1)} Z_v^{(2)}, \\ \frac{\mathrm{d}Z_v^{(2)}}{\mathrm{d}v} &= \delta Z_v^{(1)} Z_v^{(2)} - \gamma Z_v^{(2)}, \end{aligned}$$

where $Z_v^{(1)}$ and $Z_v^{(2)}$ respectively represent the prey and predator population densities, i.e. the number of individuals per unit area. The known initial values are $Z_0^{(1)} = 21$ and $Z_0^{(2)} = 9$. The components of the parameter are the prey growth rate $\alpha > 0$, predation rate $\beta > 0$, predator growth rate $\delta > 0$, predator death rate $\gamma > 0$.

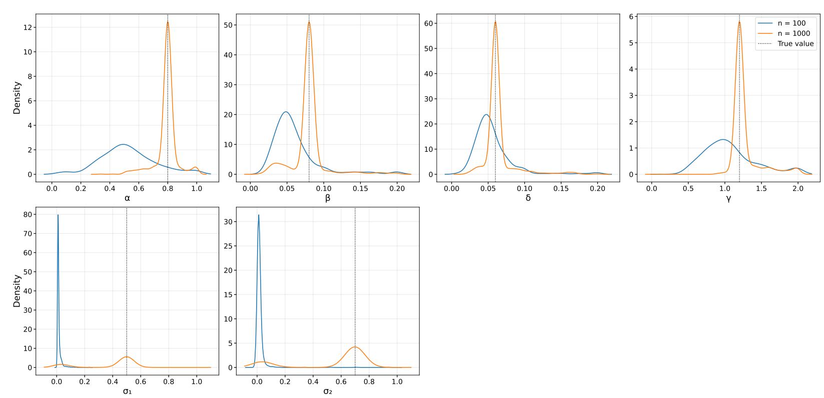
The predator-prey populations are observed through noise. For $t = 1, \dots, n$ and $j \in \{1, 2\}$, define

$$X_t^{(j)} = Z_{Tt/n}^{(j)} + \sigma^{(j)} \epsilon_t^{(j)},$$

where each $\epsilon_t^{(j)} \overset{\text{iid}}{\sim} N(0,1)$. $X_t^{(1)}$, $X_t^{(2)}$ denote the observed prey and predator populations, and $\epsilon_t^{(1)}$, $\epsilon_t^{(2)}$ represent the measurement noise with the noise level controlled by $\sigma^{(1)}$, $\sigma^{(2)}$. We aim to estimate

$$\begin{aligned} \alpha_0 &= 0.8, & \beta_0 &= 0.08, & \delta_0 &= 0.06, \\ \gamma_0 &= 1.2, & \sigma_0^{(1)} &= 0.5, & \sigma_0^{(2)} &= 0.7, \end{aligned}$$

using the rolling-window estimator.



Structural time series model. We consider a two-dimensional structural time series model with cycles, trends, and an abrupt change-point. For $u \in [0,1]$, define

$$\begin{aligned} Z_u^{(1)} &= \mu^{(1)}u + \alpha \mathbf{1}_{u \geq \tau} + \cos(2\pi\beta^{(1)}u), \\ Z_u^{(2)} &= \mu^{(2)}u - \alpha \mathbf{1}_{u \geq \tau} + \sin(2\pi\beta^{(2)}u). \end{aligned}$$

For $t = 1, \ldots, n$ and $j \in \{1, 2\}$, we observe

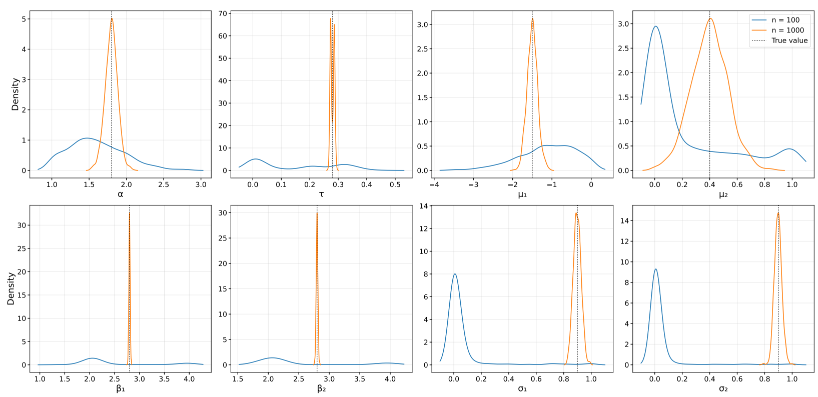
$$X_t^{(j)} = Z_{t/n}^{(j)} + \sigma^{(j)}\epsilon_t^{(j)},$$

where $\epsilon_t^{(j)} \overset{\text{iid}}{\sim} N(0,1)$ and each $\sigma^{(j)} > 0$ controls the level of the noise. We aim to estimate

$$\begin{aligned} \alpha_0 &= 1.8, & \tau_0 &= 0.28, & \mu_0^{(1)} &= -1.5, & \mu_0^{(2)} &= 0.4, \\ \beta_0^{(1)} &= 2.8, & \beta_0^{(2)} &= 2.8, & \sigma_0^{(1)} &= 0.9, & \sigma_0^{(2)} &= 0.9, \end{aligned}$$

using the rolling-window estimator.

Note that our theoretical guarantees only apply to models in which the distributions change smoothly with the parameter. The abrupt change-point in this model violates this assumption. Nevertheless, in practice, our method still reliably estimates the location of the change-point. On the other hand, our theoretical framework allows for *known* abrupt change-points and other general forms of nonstationarity. We emphasize that our theoretical framework does provide guarantees of consistency and asymptotic normality for unknown *smooth* change-points, which can be parametrized using logistic functions, for instance.



4 Prediction with Transformers via Simulation

4.1 Background

We provide a background on topics that will be discussed in this section.

Transformers and Time Series Foundation Models (TSFMs). Drawing inspiration from the remarkable success of transformer-based foundation models for language and vision tasks, many researchers are now seeking to develop foundation models for time series using transformers (Vaswani et al., 2017). *Time series foundation models* (TSFMs) are large models, with millions or even billions of parameters, that are pretrained on massive collections of real-world and synthetic time series. There has recently been a growing emphasis on training TSFMs using synthetic time series (Ansari et al., 2024, 2025; Liu et al., 2025). In fact, many TSFMs are trained *exclusively* on synthetic time series (Dooley et al., 2023; Bhethanabhotla et al., 2024; Verdenius et al., 2024; Taga et al., 2025; Oreshkin et al., 2026). The synthetic time series are often generated by *parametric* time series models with a variety of different parameter values. Many TSFMs are also trained on *convex combinations* of time series realizations; see TSMixup from Ansari et al. (2024). These rapid developments raise fundamental questions about how transformers trained on synthetic time series data are able to learn useful representations of temporal dynamics, especially in nonstationary settings.

Simulation-and-Regression. We take inspiration from the *simulation-and-regression* methods of Longstaff and Schwartz (2001) and Moussa et al. (2026). The *least squares Monte Carlo* (LSMC) method from Longstaff and Schwartz (2001) was originally developed for pricing American options. For the purposes of this discussion, we will gloss over the details involving finance. In the LSMC algorithm, simulation is used to estimate the regression function f for two random variables X, Y by drawing $X^{(r)}$, $Y^{(r)}$, $r = 1, \dots, s$, and minimizing the squared loss

$$\frac{1}{s}\sum_{r=1}^{s}(X^{(r)} - f(Y^{(r)}))^2,$$

over some function class to obtain an estimate $\hat{f}$ of the regression function f, where the simulations are drawn from the model with some selected parameter.

Similarly, the *extremum Monte Carlo* (XMC) method from Moussa et al. (2026) uses regression (e.g., random forests (Breiman, 2001)) or quantile regression (e.g., quantile regression forests (Meinshausen and Ridgeway, 2006)) to perform forecasting and smoothing for state-space models when the parameter is assumed to be known. In practice, the parameter is estimated (e.g., using simulation-based estimation) and treated as known. Consequently, the estimators in the XMC method minimize a simulation-based empirical risk with respect to some loss L (e.g., squared loss or pinball loss), where the simulations are drawn from the distribution corresponding to the estimated parameter. That is, they aim to find the function f in some function class that minimizes the Monte Carlo approximation to the risk

$$R(f) = \mathbb{E}_{P_{\hat{\theta}}}\left[L\left(X - f(Y)\right)\right],$$

for some loss L, where the expectation is with respect to the distribution $P_{\hat{\theta}}$ of (X, Y) corresponding to the estimated parameter $\hat{\theta}$. Typically, the estimate $\hat{\theta}$ is not equal to the true parameter θ_0.

4.2 State-space Modeling

We give a brief overview of forecasting and smoothing for state-space models. We emphasize that our proposed work on transformer-based time series prediction is not restricted to state-space models, or even to individual time series models. For example, we can use convex combinations of time series models, as in TSMixup from Ansari et al. (2024). Nevertheless, the following definitions and examples will be helpful to have in mind for the discussions in Section 4.4.

State-space models decompose observed time series $Y_{1:t}$ into latent states $X_{1:t}$, control inputs $Z_{1:t}$, and noise. Consider the (possibly nonstationary) state-space model for an $\mathbb{R}^{d_Y}$-valued observed process $Y_{1:n}$ and an $\mathbb{R}^{d_X}$-valued latent state process $X_{1:n}$. For times $t = 1, \ldots, n$, let

$$\begin{aligned} Y_t &= M_t(X_t, Z_t, \varepsilon_t^Y, \theta), & (\varepsilon_t^X, \varepsilon_t^Y) &\sim P_\theta(\varepsilon_t^X, \varepsilon_t^Y), \\ X_t &= S_t(X_{t-1}, Z_t, \varepsilon_t^X, \theta), & X_0 &\sim P_\theta(X_0), \end{aligned} \tag{15}$$

where M_t is the measurement function at time t, S_t is the state transition function at time t, and $\theta \in \Theta \subset \mathbb{R}^p$ is a p-dimensional parameter.

We explain the ideas of smoothing and forecasting in the context of regression for simplicity. The same ideas apply for quantile regression by replacing the conditional expectations with the corresponding conditional quantiles. In the context of state-space modeling, *smoothing* is the task of estimating the past latent states $X_{1:t}$ using the information up to time t. This can be done via simulation-based estimates of the regression functions $\mathbb{E}(X_j \mid Y_{1:t} = y_{1:t}, Z_{1:t} = z_{1:t})$ for $j = 1, \ldots, t$. *Forecasting* is the task of predicting the future observations or states multiple time-steps ahead. In the state-space setting, a future path of the control inputs must be specified, either deterministically or stochastically, or the model must be specified so that no controls are used. Given the observations $Y_{1:t}$ and (optionally) control inputs $Z_{1:t+h}$, we aim to make point forecasts via simulation-based estimates of $\mathbb{E}(X_{t+j} \mid Y_{1:t} = y_{1:t}, Z_{1:t+j} = z_{1:t+j})$ or $\mathbb{E}(Y_{t+j} \mid Y_{1:t} = y_{1:t}, Z_{1:t+j} = z_{1:t+j})$ for $j = 1, \ldots, h$.

We present some examples. We will use versions of these models in the papers that apply our proposed methods. Specifically, we use (16) to estimate a certain notion of latent infectivity in epidemiology via smoothing, and we use (17) for generating synthetic time series for forecasting as in Dooley et al. (2023).

First, consider a process with $\mathbb{R}^{d_Z}$-valued controls $Z_{1:n}$. We condition on the controls, so that we may view each control as a discretely observed function of continuous time that is suitably nice (e.g., Hölder continuous in time)

$$\begin{aligned} Y_t &\sim \exp\left(\kappa X_t + \beta^\top Z_t + \mu\right) \varepsilon_t^Y, \\ X_t &= \phi X_{t-1} + \varepsilon_t^X, \end{aligned} \tag{16}$$

for some effect sizes $\kappa \in \mathbb{R}$, $\beta \in \mathbb{R}^{d_Z}$, autoregressive coefficient $\phi \in (-1, 1)$, and intercept term $\mu \in \mathbb{R}$, with $X_0 \sim N\left(0, \frac{\sigma_X^2}{1-\phi^2}\right)$ and $(\varepsilon_t^X, \varepsilon_t^Y) \sim N(\mathbf{0}, \mathbf{\Sigma})$, where

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix},$$

for some $\sigma_X, \sigma_Y > 0$, $\rho \in [-1, 1]$. A similar model was considered by Hol and Koopman (2000). Second, consider a multiplicative trend-seasonality-noise decomposition with no controls

$$\begin{aligned} Y_t &= \mu(t, \theta^\mu)\exp\left(\kappa X_t - \frac{1}{2}\kappa^2 \frac{\sigma_X^2}{1-\phi^2}\right) \varepsilon_t^Y, \\ X_t &= \phi X_{t-1} + \varepsilon_t^X, \end{aligned} \tag{17}$$

for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N\left(0, \frac{\sigma_X^2}{1-\phi^2}\right)$ and $(\varepsilon_t^X, \varepsilon_t^Y) \sim N(\mathbf{0}, \mathbf{\Sigma})$, where

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix},$$

for some $\sigma_X, \sigma_Y > 0$, $\rho \in [-1, 1]$, and

$$\mu(t, \theta^\mu) = \text{seasonality}\left(t, \theta^{Se}\right) \ \text{trend}\left(t, \theta^{Tr}\right),$$

decomposes the mean function into a product of seasonality and trend functions with structural parameters $\theta^\mu = \text{Vec}(\theta^{Se}, \theta^{Tr})$. For pre-training their prior-data fitted network for time series forecasting, Dooley et al. (2023) considers a seasonality function based on a linear combination of sines and cosines (with weekly, monthly, and yearly periods), and a trend function based on multiplying a linear trend and exponential trend.

We make a few remarks about the exponentiated autoregressive process from (17). Recall that the stationary distribution of X_t is $N\left(0, \frac{\sigma_X^2}{1-\phi^2}\right)$, so the stationary distribution of $\exp(\kappa X_t)$ is log-normal with mean $\exp(\frac{1}{2}\kappa^2 \frac{\sigma_X^2}{1-\phi^2})$.

Hence, the stationary distribution of $\exp(\kappa X_t - \frac{1}{2}\kappa^2 \frac{\sigma_X^2}{1-\phi^2})$ is log-normal with mean $\exp(0) = 1$, and the (local) expectation of the observation Y_t at time t is the mean function $\mu(t, \theta^\mu)$. Therefore, we can interpret this multiplicative decomposition with the exponentiated autoregressive process as a way to stochastically stretch and compress the deterministic mean function, in a manner that is compatible with the framework of locally stationary time series, in the sense of Dahlhaus et al. (2019). When using this model in the context of training transformers for forecasting, this model can be seen as an extension of the generative model used by Dooley et al. (2023).

To estimate the parameters of these models, we use the rolling-window estimator for nonstationary time series from Section 3. It is straightforward to show that these models satisfy the sufficient conditions for our estimator to be consistent and asymptotically normal (i.e. within a certain framework for nonstationary time series, which builds on Dahlhaus et al. (2019); Mies (2023)). Based on Lemma 2.4 and Corollary 2.2 in Kokoszka et al. (2025), it is straightforward to prove that there is a decay of the physical dependence measure for these nonstationary state-space models models with exponentiated autoregressive processes.

4.3 Confidence Distributions

Our proposed method can be viewed as a frequentist, likelihood-free analogue of the Laplace approximation in Bayesian statistics. Specifically, when using the Laplace approximation to obtain an approximate posterior predictive distribution. Recall that the Laplace approximation uses a Gaussian distribution centered at the maximum a posteriori (MAP) estimate with covariance given by the inverse observed Fisher information to approximate the posterior distribution. The Bernstein–von Mises theorem is used to justify the Laplace approximation. In contrast, to justify our approach, we use asymptotic theory for likelihood-free parameter estimators for parametric time series models (e.g., Section 3), nonstationary nonlinear time series theory (Mies, 2023), and the theory of confidence distributions.

In general, our frequentist procedure can be used with any *confidence distribution* over $\Theta \subset \mathbb{R}^p$ which expresses our uncertainty about the unknown parameter $\theta_0 \in \mathbb{R}^p$. Ideally, this distribution is easy to simulate from. We give the definition of confidence distributions for the $p = 1$ case, and the generalization to the $p \in \mathbb{N}$ case. In the definition, $\Theta \subset \mathbb{R}^p$ is the parameter space and $\mathcal{X}_n$ is the sample space corresponding to the data $X_{1:n}$.

Definition 1 (Confidence Distribution, $p = 1$, (Xie and Singh, 2013))**.** *A* *function* $\hat{\mathrm{CD}}_n(\cdot) = \hat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ *on* $\mathcal{X}_n \times \Theta \to [0,1]$ *is called a confidence distribution (CD) for a parameter* θ*, if:*

1. *For each given* $X_{1:n}$, $\hat{\mathrm{CD}}_n(\cdot) = \hat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ *is a cumulative distribution function on* Θ*.*

2. *At the true parameter value* $\theta = \theta_0$, $\hat{\mathrm{CD}}_n(\theta_0) \equiv \hat{\mathrm{CD}}_n(X_{1:n}, \theta_0)$*, as a function of the sample* $X_{1:n}$*, follows the uniform distribution* $U[0,1]$*.*

Also, the function $\hat{\mathrm{CD}}_n(\cdot)$ *is an asymptotic confidence distribution (aCD), if the* $U[0,1]$ *requirement is true only asymptotically.*

The definition of confidence distributions for the general $p \in \mathbb{N}$ case involves depth functions. As discussed in Liu et al. (1999), a data depth is a way of measuring how deep or central a point $x \in \mathbb{R}^d$ is with respect to a probability distribution P on $\mathbb{R}^d$, $d \in \mathbb{N}$, or with respect to a given data cloud $X_{1:n}$. For example, the Mahalanobis depth (D_M) (Mahalanobis, 1936) at $x \in \mathbb{R}^d$ with respect to P is defined as

$$D_M(P, x) = \frac{1}{1 + (x - \mu_P)^\top \Sigma_P^{-1}(x - \mu_P)},$$

where μ_P and Σ_P are the mean vector and the dispersion matrix of P, respectively. We obtain the sample version of D_M by replacing μ_P and Σ_P by the corresponding sample estimates $\hat{\mu}_P$ and $\hat{\Sigma}_P$. See Liu et al. (1999) for more examples of data depth. In general, a point with a larger depth value (i.e. closer to 1) means that it is closer to the center of the distribution or more central in a data cloud, while points with smaller depth values (i.e. closer to 0) are farther away from the center, so data depth is a center-outward ordering of multivariate observations.

Definition 2 (Depth Confidence Distribution, $p \in \mathbb{N}$, (Liu et al., 2022))**.** *A function* $\hat{\mathrm{CD}}_n(\cdot) \equiv \hat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ *on* $\Theta \subset \mathbb{R}^p$ *is called a depth confidence distribution (depth-CD) associated with depth function* D *for a vector-valued parameter* θ*, if:*

1. *It is a distribution function on the parameter space* Θ *for any fixed sample set* $X_{1:n}$*.*

2. *The* $(1-\alpha)$ *"central region" of the distribution* $\hat{\mathrm{CD}}_n(\cdot)$, $\mathcal{R}_{1-\alpha}(\hat{\mathrm{CD}}_n) = \{\vartheta \in \Theta : C_{\hat{\mathrm{CD}}_n}(\vartheta) \geq \alpha\}$*, is a confidence region for* θ *with a coverage probability of* $(1-\alpha)$*. Here, the centrality function associated with depth* D *and confidence distribution* $\hat{\mathrm{CD}}_n(\cdot)$*, that is,* $C_{\hat{\mathrm{CD}}_n}(\vartheta)$*, is also referred to as a depth confidence curve (depth-CV).*

If the statements in (2) hold only asymptotically, then we refer to $\hat{\mathrm{CD}}_n$ *and* $C_{\hat{\mathrm{CD}}_n}$ *as asymptotic depth-CD and asymptotic depth-CV, respectively.*

In this work, we focus on asymptotic Gaussian depth confidence distributions of the form $\hat{\mathrm{CD}}_n = N(\hat{\theta}, \hat{\Sigma}/n)$ for a multi-dimensional parameter $\theta \in \mathbb{R}^p$, $p \in \mathbb{N}$. See Example 3 in Liu et al. (2022) for an illustration with a bivariate normal distribution. We justify this Gaussianity through the asymptotic normality of the parameter estimator (e.g., Section 3), nonstationary nonlinear time series theory (Mies, 2023), and the following result from Liu et al. (2022) (Theorem 2 therein).

Theorem 3 (Depth-CDs from Pivot Statistics (Liu et al., 2022))**.** *Assume that* $A_n(X_{1:n})$ *is a nonsingular matrix such that* $A_n(X_{1:n})(\hat{\theta} - \theta)$ *follows a distribution* Q_n *that is free of all unknown parameters. Also assume that* η_n *is a random vector, independent of the sample* $X_{1:n}$*, following the distribution* Q_n*. Then, conditional on* $X_{1:n}$*, the distribution function of* $(\hat{\theta} - A_n(X_{1:n})^{-1}\eta_n)$ *is a depth-CD for* θ*, under the following conditions:*

1. *The depth* D *is affine-invariant.*
2. *The depth contours* $\{\eta_n : D_{Q_n}(\eta_n) = \epsilon\}$ *all have probability zero with respect to* Q_n *for any* $\epsilon > 0$.

Alternatively, a parametric bootstrap approach can be used (Efron, 1979; Efron and Tibshirani, 1994). To use a parametric bootstrap approach, we would estimate the parameter on the observed data, simulate from the model with the estimated parameter, then re-estimate the parameter for each simulated realization. The sampling distribution of the re-estimates is used as the depth confidence distribution. Theorem 1 in Liu et al. (2022) shows that this bootstrap distribution is an asymptotic depth confidence distribution under certain regularity conditions. For more information about the connections between bootstrap distributions and confidence distributions, see Example 3 in Xie and Singh (2013) and the related discussions in Efron (1998).

We opt for an asymptotic Gaussian depth confidence distribution, rather than a bootstrap depth confidence distribution, to improve the computational efficiency of our method. This computational efficiency is particularly important, because our proposed method is intended to be applicable to models with intractable likelihoods, where parameter estimation typically relies on simulation-based procedures that are substantially more computationally demanding than methods like maximum likelihood estimation. Therefore, our procedure shares some of the motivations behind the Krinsky and Robb (1986) method.

4.4 Proposed Work

Motivated by the topics discussed in Section 4.1, we study simulation-based time series prediction with transformers. Specifically, we focus on two time series prediction tasks: (1) forecasting future observations and latent states, and (2) smoothing past latent states. Broadly speaking, our goals are to: (1) understand the statistical foundations of simulation-based approaches to training transformers from scratch and post-hoc training of pre-trained transformers (i.e. fine-tuning), and (2) identify principles that enable transformers to achieve satisfactory performance on time series prediction tasks.

Our method inherits the computational benefits of sequence-to-sequence modeling. In the context of smoothing, our proposed method only requires training one sequence-to-sequence model, as opposed to separate models for each time as in Moussa et al. (2026). Similarly, we only require one sequence-to-sequence model to obtain multi-step forecasts.

We propose a transformer-based simulation-and-regression framework that incorporates uncertainty about the true parameter. We suggest minimizing a simulation-based empirical risk using a sequence-to-sequence loss, where the simulations are generated by first drawing a parameter from an estimated *confidence distribution* over the parameter space $\Theta \subset \mathbb{R}^p$ (see Definition 2) and then simulating a time series from the model at that parameter. That is, we aim to find the sequence-to-sequence function f in some function class (i.e., the class of transformer networks) that minimizes the Monte Carlo approximation to the risk

$$\begin{aligned} R_{\hat{P}_{\text{mix}}}(f) &= \mathbb{E}_{\hat{P}_{\text{mix}}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right] \\ &= \mathbb{E}_{\theta\sim\hat{\text{CD}}_n}\left[\mathbb{E}_{P_\theta}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right]\right], \end{aligned} \tag{18}$$

where the second equality holds by Fubini's theorem. Here, L is some loss,

$$\hat{P}_{\text{mix}} = \int_\Theta P_\theta d\hat{\text{CD}}(\theta),$$

is the mixture distribution, $\hat{\text{CD}}_n$ is the estimated confidence distribution over Θ, P_θ is the distribution of the sequences corresponding to θ, $\mathcal{T}$ is a set of times, and $[f(Y)]_t$ is the sequence-to-sequence function at time t. The details of the procedure are presented in Section 4.5.

In contrast, the simulation-and-regression methods from Longstaff and Schwartz (2001) and Moussa et al. (2026), discussed in Section 4.1, only simulate from the model at the estimated parameter. The corresponding sequence-to-sequence version of the XMC method from Moussa et al. (2026) would minimize the Monte Carlo approximation to the risk

$$R_{P_{\hat{\theta}}}(f) = \mathbb{E}_{P_{\hat{\theta}}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right], \tag{19}$$

over some sequence-to-sequence function class. Here, L is some loss, $P_{\hat{\theta}}$ is the distribution of the sequences corresponding to the estimated parameter $\hat{\theta}$, $\mathcal{T}$ is a set of times, and $[f(Y)]_t$ is the sequence-to-sequence function at time t.

Question 1. Under what conditions does accounting for parameter uncertainty via our procedure typically achieve better performance than just using the parameter point estimate? Specifically, we aim to explain when

$$\mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{\hat{P}_{\mathrm{mix}}}(Y)]_t\right)\right] < \mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\hat{\theta}}}(Y)]_t\right)\right],$$

where both expectations are with respect to the distribution P_{θ_0} of the sequences corresponding to the unknown true parameter θ_0, $f^*_{\hat{P}_{\mathrm{mix}}}$ is the minimizer in the function class of $R_{\hat{P}_{\mathrm{mix}}}(f)$ from (18), and $f^*_{P_{\hat{\theta}}}$ is the minimizer in the function class of $R_{P_{\hat{\theta}}}(f)$ from (19).

When the asymptotic variance of the parameter estimator is “relatively small”, there is more concentration of the simulation budget in regions of the parameter space near the parameter estimate. On the other hand, when the asymptotic variance of the parameter estimator is “relatively large”, there is less concentration of the simulation budget in regions of the parameter space near the parameter estimate. Hence, roughly speaking, the extremes are a point mass at the parameter estimate and a uniform distribution over the parameter space.

In this sense, our method can be situated between two existing paradigms in simulation-based time series prediction: (1) simulation-and-regression for time series using parameter point estimates as in Moussa et al. (2026), and (2) time series foundation models pre-trained on synthetic time series as in

Dooley et al. (2023); Bhethanabhotla et al. (2024); Verdenius et al. (2024); Taga et al. (2025); Oreshkin et al. (2026).

> **Question 2.** When used with an untrained transformer, our procedure can be viewed as a *robust simulation-and-regression method*, as it accounts for parameter uncertainty during training. When used with a pre-trained transformer, our method produces a *contextualized time series foundation model*, as it performs post-hoc training (i.e. fine-tuning) with relevant synthetic time series. Under what conditions does a time series foundation model contextualized with our procedure typically outperform a transformer trained from scratch?

Intuitively, this occurs when the following conditions hold. First, there is large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise. Second, the distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions. Nevertheless, a rigorous characterization is desirable.

Along the way, we will show that the estimators obtained by the proposed procedures are consistent as the sample size n and number of simulations s tend to infinity. The convergence rate is determined by the parameter estimator (see Section 3) and sequence-to-sequence function estimator (see Takakura and Suzuki (2023); Kim et al. (2024)). We will study the finite-sample performance of the proposed procedure and compare with existing approaches.

Lastly, we instantiate our framework in the context of sequence-to-sequence quantile regression. In this case, the loss function L is chosen to be the pinball loss. The following questions, inspired by the work of Adler et al. (2025), are pertinent to transformer models that output multiple conditional quantiles (i.e. quantile prediction heads). Typically, the set of quantiles $\tau \in \{0.1, 0.2, \ldots, 0.9\}$ is used. For example, see Das et al. (2024); Auer et al. (2025); Wang et al. (2025).

Question 3. Under what conditions: (i) Does accounting for parameter uncertainty via our procedure improve the calibration of transformers? (ii) Does our procedure result in well-calibrated transformers, when applied to untrained transformers? (iii) Does our procedure improve the calibration of time series foundation models, when applied to pre-trained transformers?

4.5 Algorithms

The proposed forecasting and smoothing methods are presented below. Note that our proposed multi-step forecasting method can be paired with conformal prediction methods for multi-step forecasting of nonstationary time series; see Wang and Hyndman (2024).

The proposed procedures utilize an estimator $\hat{\theta}$ of the true parameter θ_0 of a (possibly nonstationary) parametric time series model. The estimator is assumed to be consistent and asymptotically normal. For example, the estimators from Section 3 can be used. For the cross-validation scheme, we introduce an aggregation function $\mathrm{Agg}(\cdot)$ for more flexibility. For example, $\mathrm{Agg}(\cdot)$ can be taken to be the average over the K folds, or just the error in the validation fold if using an 80/20 train/validation split. As usual, for the loss function L, the squared loss is used for regression and the pinball loss is used for quantile regression.

Forecasting with Transformers.

- **Inputs:** Observations $Y_{1:n}^{\mathrm{obs}}$, state-space model (15), number of simulations s, number of folds K for cross-validation, proportion of simulations in each fold $c_1, \ldots, c_K$, cross-validation error aggregation function $\mathrm{Agg}(\cdot)$, estimator $\hat{\theta}$, estimation method for asymptotic covariance of $\hat{\theta}$, forecasting horizon h, and loss function L.

- Obtain estimate $\hat{\theta}$ of the true parameter θ_0.

- Obtain estimate $\hat{\Sigma}$ of the asymptotic covariance Σ of estimator $\hat{\theta}$.

- For each $r \in [s]$, draw $\theta^{(r)} \sim N(\hat{\theta}, \hat{\Sigma}/n)$ from the confidence distribution, and simulate from the model (15),

$$Y_{1:n+h}^{(r)} \sim P_{n+h,\theta^{(r)}}.$$

- Denote the r-th sequence-to-sequence loss by

$$\ell_r(f) = \sum_{t=n+1}^{n+h} L(Y_t^{(r)} - [f(Y_{1:n}^{(r)})]_t),$$

where $[f(Y_{1:n}^{(r)})]_t$ is the r-th sequence-to-sequence prediction at time t.

- Partition the simulation indices $[s]$ into K folds $F_1, \ldots, F_K$, where,

$$F_i = \{r \in [s] : \sum_{j=1}^{i-1} c_j s < r \le \sum_{j=1}^{i} c_j s\}, \quad i = 1, \ldots, K.$$

- For each $k \in [K]$ and each value of the tuning parameter $\lambda \in \Lambda$, obtain the minimizer $\hat{f}_\lambda^{-k}$ of the average loss

$$\frac{1}{|F_{-k}|} \sum_{r \in F_{-k}} \ell_r(f),$$

over the function space of transformers, where $F_{-k} = \bigcup_{j \in [K] \setminus \{k\}} F_j$.

- Record the corresponding error on the k-th validation set

$$e_k(\lambda) = \frac{1}{|F_k|} \sum_{r \in F_k} \ell_r(\hat{f}_\lambda^{-k}).$$

- Compute the aggregated error over the K folds

$$\mathrm{CV}(\lambda) = \mathrm{Agg}(e_1(\lambda), \ldots, e_K(\lambda)).$$

- Find the value λ^* of the tuning parameter λ that minimizes $\mathrm{CV}(\lambda)$.
- Obtain the estimate $\hat{f}$ by minimizing the average loss using all s simulations with the selected tuning parameter λ^* to

$$\frac{1}{s} \sum_{r \in [s]} \ell_r(f).$$

- Apply the estimated function $\hat{f}$ to make the multi-step forecast of the future observations

$$\hat{Y}_{n+1:n+h} = \hat{f}(Y_{1:n}^{\mathrm{obs}}).$$

Smoothing with Transformers.

- **Inputs:** Observations $Y_{1:n}^{\text{obs}}$, state-space model (15), number of simulations s, number of folds K for cross-validation, proportion of simulations in each fold $c_1, \ldots, c_K$, cross-validation error aggregation function $\text{Agg}(\cdot)$, estimator $\hat{\theta}$, estimation method for asymptotic covariance of $\hat{\theta}$, and loss function L.
- Obtain estimate $\hat{\theta}$ of the true parameter θ_0.
- Obtain estimate $\hat{\Sigma}$ of the asymptotic covariance Σ of estimator $\hat{\theta}$.
- For each $r \in [s]$, draw $\theta^{(r)} \sim N(\hat{\theta}, \hat{\Sigma}/n)$ from the confidence distribution, and simulate from the model (15),
$$X_{1:n}^{(r)}, Y_{1:n}^{(r)} \sim P_{n,\theta^{(r)}}.$$
- Denote the r-th sequence-to-sequence loss by
$$\ell_r(f) = \sum_{t \in [n]} L(X_t^{(r)} - [f(Y_{1:n}^{(r)})]_t),$$
where $[f(Y_{1:n}^{(r)})]_t$ is the r-th sequence-to-sequence prediction at time t.
- Partition the simulation indices $[s]$ into K folds $F_1, \ldots, F_K$, where,
$$F_i = \{r \in [s] : \sum_{j=1}^{i-1} c_j s < r \leq \sum_{j=1}^{i} c_j s\}, \quad i = 1, \ldots, K.$$
- For each $k \in [K]$ and each value of the tuning parameter $\lambda \in \Lambda$, obtain the minimizer $\hat{f}_\lambda^{-k}$ of the average loss
$$\frac{1}{|F_{-k}|} \sum_{r \in F_{-k}} \ell_r(f),$$
over the function space of transformers, where $F_{-k} = \bigcup_{j \in [K] \setminus \{k\}} F_j$.
- Record the corresponding error on the k-th validation set
$$e_k(\lambda) = \frac{1}{|F_k|} \sum_{r \in F_k} \ell_r(\hat{f}_\lambda^{-k}).$$

- Compute the aggregated error over the K folds

$$\mathrm{CV}(\lambda) = \mathrm{Agg}(e_1(\lambda), \ldots, e_K(\lambda)).$$

- Find the value λ^* of the tuning parameter λ that minimizes $\mathrm{CV}(\lambda)$.
- Obtain the estimate $\hat{f}$ by minimizing the average loss using all s simulations with the selected tuning parameter λ^* to

$$\frac{1}{s} \sum_{r \in [s]} \ell_r(f).$$

- Apply the estimated function $\hat{f}$ to predict the latent states

$$\hat{X}_{1:n} = \hat{f}(Y_{1:n}^{\mathrm{obs}}).$$

5 Timeline

Step 1a: Proposal. I plan to propose my thesis on April 3rd, 2026.

Step 1b: Revise and Submit. I will submit the revision of the conditional independence testing paper corresponding to Section 2, which has received a revise decision from the journal, by May 3rd, 2026. Soon afterwards, I will submit the simulation-based parameter estimation paper corresponding to Section 3.

Step 2a: Statistical Foundations of Time Series Foundation Models Trained on Synthetic Data. The main paper will answer Questions 1 and 2 from Section 4.4 using all the material from Section 4. This is a theory and methods paper, so I plan to work on it from May 2026 to March 2027. While working on the main paper, I will submit the following two smaller papers.

Step 2b: Calibrating Time Series Foundation Models by Accounting for Parameter Uncertainty. Once the forecasting method from Section 4.5 has been implemented, I will answer Question 3 from Section 4.4 by writing a paper about the calibration of quantile forecasts using a version of the model from (17). This paper will involve training (relatively small) transformers from scratch and fine-tuning pre-trained transformers, so I expect it to take about six weeks.

Step 2c: Estimating Population Infectivity with Transformers using Viral Load Distributions, Variants, Vaccinations, and Mobility. Once the smoothing method from Section 4.5 has been implemented, I will apply it to epidemiology. Specifically, I will estimate a set of quantiles of a latent infectivity process using a version of the model from (16). This is an applied paper, and I have already prepared the data (variant relative frequency, vaccine coverage, mobility metrics, case growth rates, and Ct values from RT-qPCR tests), so I expect this to take about three weeks.

Step 3: Defense. I plan to defend my thesis before the spring defense deadline set by the Department, April 15, 2027, and submit the paperwork by May 1, 2027.

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